My Self-Directed MSc in *Cognitive Science*

A world-class curriculum synthesised from MIT, CMU, Stanford, Edinburgh, UCSD, Indiana and Sussex.

DURATION	STRUCTURE	OUTPUT	PACE
80 weeks	**5** × 21	**21** artifacts	**12–14**
~18 months, self-paced	Semesters × modules	Publishable quality	hrs / week
			Alongside full-time work

§ 01

Progress

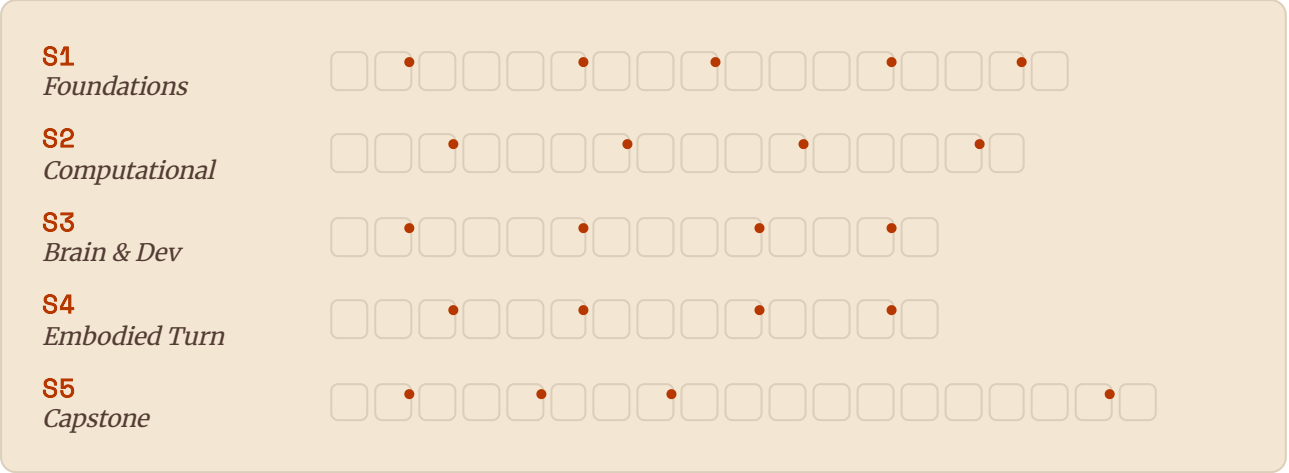
0%
COMPLETE

WEEKS
0/80

CURRENT
Not started

ARTIFACTS
0/21

NEXT ARTIFACT
Week 3 · Paradigm Map Essay



WEEK · COMPLETED · IN PROGRESS · AWAITING LINK · ARTIFACT DUE

§ 02 How this works

EACH MODULE CONTAINS

Core readings (textbooks + primary papers), lectures (free online courses and video series), and an artifact — the same kind of submission a masters student would turn in.

PACING

One module every 3–4 weeks. Heavier modules (10 ECTS equivalent) get 4 weeks; lighter ones get 3. The capstone gets 10 weeks. Expect the 80-week schedule to slip to 90–100 weeks in practice — that is normal and accounted for.

ASSESSMENT

Every artifact is written to publishable or submission-ready quality. Share with peers, online communities (LessWrong, PhilPapers discussions, r/CogSci), or use AI-assisted review. For the capstone: budgeted paid reads from two PhD students in the field — the single change that most narrows the gap to an accredited programme.

Weekly time budget

Reading	5–6 hrs	Textbooks + primary papers
Lectures	2–3 hrs	Free university video series
Coding / stats	3–4 hrs	Modeling + reanalyses
Writing	2–4 hrs	Notes, drafts, artifacts

1–2 hours on weekday evenings for reading and lectures; 4–6 hours on weekends for writing, coding, and deep thinking.

§ 03 **Curriculum**

SEMESTER 1 · WEEKS 1–17

Foundations

Build the conceptual map of the field — and the methods spine that will carry the rest of the degree.

MODULE 1.1 · WEEKS 1–3

Introduction to Cognitive Science

> *What is cognitive science? What are the competing paradigms? Where did they come from?*

W01 The Landscape

- **READ** Bermúdez Ch 1–5 — history of cognitive science, classical computationalism, physical symbol systems
- **WATCH** Berkeley COGSCI 1 — Lectures 1–6 (field overview, history, methods)
- **PAPER** Turing (1950) — Computing Machinery and Intelligence
- **NOTE** *Start a running glossary of key terms and positions. You'll reference it all programme.*

W02 The Paradigms

- **READ** Bermúdez Ch 6–10 — connectionism, dynamical systems, Bayesian approaches
- **WATCH** Berkeley COGSCI 1 — Lectures 7–14
- **PAPER** McCulloch & Pitts (1943) · Newell & Simon (1956)
- **READ** Skim: Dawson Mind, Body, World Ch 1–3 — second perspective on the same territory

W03
12–14 hrs

Synthesis + Artifact

READ Bermúdez Ch 11–14 — embodied cognition, consciousness, the future

WATCH Berkeley COGSCI 1 — Lectures 15–25

PAPER Miller (1956) — The Magical Number Seven

ARTIFACT DUE **Paradigm Map Essay**
Written · 3,000 words

MODULE 1.2 · WEEKS 4–7

Philosophy of Mind

> *What is a mind? What is consciousness? Can machines think?*

W04

The Mind–Body Problem

READ Kim Philosophy of Mind Ch 1–4 — substance dualism, behaviorism, mind-brain identity

WATCH Edinburgh Coursera "Philosophy of Cognitive Sciences" — Weeks 1–2

PAPER Putnam (1967) — The Nature of Mental States

W05

Functionalism and Computation

READ Kim Ch 5–7 — functionalism, machine functionalism, mental causation

PAPER Searle (1980) — Minds, Brains, and Programs + peer commentary

PAPER Fodor (1983) — Modularity of Mind Part I

WATCH Coursera — Weeks 3–4

W06 Consciousness and Qualia

- READ Kim Ch 8–10 — consciousness, qualia, the explanatory gap
- PAPER Nagel (1974) — What Is It Like to Be a Bat?
- PAPER Jackson (1982) — Epiphenomenal Qualia
- WATCH Coursera — Weeks 5–6
- WRITE Start outlining your essay — thesis + argument structure

W07 Artifact
14–16 hrs

- READ Reread key sections of Searle, Fodor, Putnam for your argument

ARTIFACT DUE **Critical Essay on the Computational Theory of Mind**
Written · 4,000 words

MODULE 1.3 · WEEKS 8–10

Cognitive Psychology

> *How do humans actually perceive, attend, remember, decide, and reason?*

W08 Perception and Attention

- READ Gazzaniga, Ivry & Mangun — Cognitive Neuroscience Ch 5–7 (perception, attention)
- PAPER Shepard & Metzler (1971) — Mental Rotation
- PAPER Sperry (1968) — Hemisphere Deconnection
- NOTE *Switched from Goldstein to Gazzaniga — the standard MSc-level text; integrates neuroscience from the start.*

W09 Memory, Reasoning, and Decision-Making

READ Gazzaniga et al. Ch 9 (memory), Ch 12 (decision & reasoning), Ch 13 (problem-solving)

PAPER Rosch & Mervis (1975) — Family Resemblances

PAPER Tversky & Kahneman (1974) — Heuristics and Biases

PAPER Wason (1968) — Reasoning About a Rule

DO Find the OSF pre-registration template (osf.io). Study 2–3 published pre-registrations in your target area.

W10 Artifact

10–12 hrs

WRITE Pick your phenomenon: anchoring, Stroop interference, false memory formation, etc.

WRITE Write: background, hypothesis, method, predicted results, analysis plan

NOTE *This pre-reg is a first draft — you'll tighten the stats plan after Module 1.5.*

ARTIFACT DUE **Experiment Design Document (pre-registration style)**

Written · ~2,000 words

MODULE 1.4 · WEEKS 11–14

Neuroscience & Neuroimaging Methods

> *How does the brain give rise to cognition, and by what measurements do we actually probe it?*

W11 Neural Signaling

- READ Bear, Connors & Paradiso — Neuroscience: Exploring the Brain Ch 2–4 (neurons, resting/action potential, synapses)
- WATCH MIT 9.13 The Human Brain — Lectures 1–6 (Kanwisher: intro, methods, face perception)
- NOTE *The most unfamiliar territory if you're coming from CS. Go slow. Take notes on mechanisms.*

W12 Sensory Systems

- READ Bear et al. Ch 6–10 — somatic sensation, auditory system, visual system in detail
- WATCH MIT 9.13 — Lectures 7–12 (visual system, object recognition, attention)
- PAPER Lettvin et al. (1959) — What the Frog's Eye Tells the Frog's Brain

W13 Higher Cognition, Cortex, and Cognitive Neuroscience

- READ Bear et al. Ch 12 (cortical organization), Ch 24–25 (memory systems)
- READ Gazzaniga, Ivry & Mangun Ch 2–3 — methods overview, anatomy for cog neuro
- WATCH MIT 9.13 — Lectures 13–18 (navigation, memory, language)
- READ Churchland Neurophilosophy — Ch 1–2 (bridging neuroscience and philosophy)

W14 — Neuroimaging Methods + Artifact

12–14 hrs

- **READ** Huettel, Song & McCarthy — Functional Magnetic Resonance Imaging Ch 1–5 (BOLD signal, GLM, experimental design)
- **READ** Luck — An Introduction to the ERP Technique Ch 1–3 (N400, P300, MMN)
- **PAPER** Kutas & Federmeier (2011) — Thirty years and counting: Finding meaning in the N400
- **READ** Skim: Hari & Puce — MEG-EEG Primer Ch 1–2; skim eye-tracking + pupillometry primer
- **DO** Pick one fMRI and one ERP paper from your area; annotate the methods sections end-to-end.
- **WATCH** MIT 9.13 — Lectures 19–24 (emotion, consciousness, social cognition)
- **NOTE** *The artifact is now methods-aware — each region is annotated with how we actually measured it.*

ARTIFACT DUE **Brain–Cognition–Methods Map — annotated diagram linking brain structures to cognitive functions AND to the imaging modality that licenses each claim**

Visual · Annotated diagram + ~1,500 word methods commentary

MODULE 1.5 · WEEKS 15–17

Research Methods & Statistics

> *How does the field produce credible knowledge after the replication crisis? This is the methodology spine that every later artifact leans on.*

W15 The Replication Crisis and Open Science

- PAPER Simmons, Nelson & Simonsohn (2011) — False-Positive Psychology
- PAPER Open Science Collaboration (2015) — Estimating the Reproducibility of Psychological Science
- PAPER Munafò et al. (2017) — A Manifesto for Reproducible Science
- PAPER Gelman & Loken (2013) — The Garden of Forking Paths
- DO Study two registered reports in detail; note how every degree of freedom was locked down before data collection.

W16 Bayesian Data Analysis

- READ McElreath — Statistical Rethinking Ch 1–5 (Bayesian updating, small worlds vs large worlds, geocentric models, multivariate regression)
- WATCH McElreath — Statistical Rethinking lectures 1–5 (YouTube)
- CODE Work through the Rethinking exercises in PyMC or the rethinking R package — reproduce Ch 4 & 5 figures from raw.

W17
14–16 hrs

Mixed-Effects, Multiverse, and Artifact

READ McElreath — Statistical Rethinking Ch 13 (multilevel models)

PAPER Barr, Levy, Scheepers & Tily (2013) — Random effects structure for confirmatory hypothesis testing: Keep it maximal

PAPER Baayen, Davidson & Bates (2008) — Mixed-effects modeling with crossed random effects

PAPER Steegen, Tuerlinckx, Gelman & Vanpaemel (2016) — Increasing transparency through a multiverse analysis

PAPER Wagenmakers et al. (2011) — Why psychologists must change the way they analyze their data

NOTE *This module is a spine, not an artifact factory. You will revisit it in every subsequent modelling module.*

ARTIFACT DUE **Reanalysis of a Published Dataset — Bayesian hierarchical model + multiverse analysis + write-up with pre-registered analytic choices**

Code + Report · 3,500 words + code + figures

SEMESTER 2 · WEEKS 18–33

Computational Core

Learn to model the mind computationally — with real statistics under every fit.

MODULE 2.1 · WEEKS 18–21

Computational Cognitive Science

> *How do we build formal, testable models of cognitive processes?*

W18 Modeling Foundations

- READ Farrell & Lewandowsky Ch 1–4 — why model?, the modeling cycle, parameter estimation, model comparison
- READ Marr Vision Ch 1 — the three levels of analysis (computational, algorithmic, implementational)
- WATCH MIT Brains, Minds, Machines — Tenenbaum lectures (first 3)

W19 Model Fitting and Evaluation

- READ Farrell & Lewandowsky Ch 5–8 — maximum likelihood, Bayesian estimation, model selection
- WATCH MIT Brains, Minds, Machines — Kanwisher + Poggio lectures
- CODE Set up your Python environment for cognitive modeling (NumPy, SciPy, matplotlib, Jupyter, PyMC)

W20 Building Your First Model

- READ Farrell & Lewandowsky Ch 9–12 — specific model families, hierarchical models
- WATCH MIT OCW 9.66J — browse lecture notes, pick 2–3 topics matching your chosen phenomenon
- CODE Start implementing your chosen cognitive phenomenon at the computational level (what problem is being solved?)

W21
14–16 hrs

Artifact

CODE Complete algorithmic and implementational level analyses

NOTE *Apply the Module 1.5 spine: report posterior distributions over parameters, not point estimates.*

ARTIFACT DUE **Three-Level Model Analysis**
Written + Code · 3,000 words + code

MODULE 2.2 · WEEKS 22–25

Bayesian Cognitive Science

> *The mind as a rational inference engine — probabilistic models of learning, reasoning, and perception.*

W22

Bayesian Foundations

READ Griffiths, Chater & Tenenbaum Ch 1–4 — Bayes' theorem, priors, likelihood, posterior inference

READ Princeton Bayesian Reading List — pick 3 foundational papers

CODE Implement Bayes' theorem from scratch. Solve toy problems (coin flipping, disease diagnosis).

W23

Bayesian Models of Cognition

READ Griffiths et al. Ch 5–8 — concept learning, causal reasoning, categorization

READ Lee & Wagenmakers — Bayesian Cognitive Modeling Ch 1–5

PAPER Tenenbaum (1999) — Bayesian modeling of human concept learning (the number game)

CODE Implement a Bayesian concept learning model targeting Tenenbaum's number game

W24 Advanced Topics + Data

READ Griffiths et al. Ch 9–12 — language, perception, development

PAPER Nicenboim & Vasishth (2016) — Statistical methods for linguistic research: Foundational Bayesian ideas

CODE Find human behavioral data from a published study. Fit your hierarchical Bayesian model with a full random-effects structure. Compare predictions to data.

W25 Artifact
14–16 hrs

CODE Finalize model, generate posterior predictive checks, generate comparison plots

NOTE *Compute BFs or information criteria (LOO / WAIC) per Module 1.5.*

ARTIFACT DUE **Bayesian Cognitive Model Report**
Code + Report · 3,000 words + Python code + figures

MODULE 2.3 · WEEKS 26–29

Connectionism and Neural Networks

> *Parallel distributed processing — how networks of simple units give rise to complex cognition.*

W26 PDP Foundations

READ O'Reilly & Munakata (free online) Ch 1–3 — neuroscience foundations, learning mechanisms, networks

PAPER Rumelhart, Hinton & Williams (1986) — Back-Propagation

PAPER Rosenblatt (1958) — The Perceptron

CODE Implement a simple feedforward network from scratch in Python (no frameworks). Train on XOR.

W27 Cognitive Models with Networks

- READ O'Reilly & Munakata Ch 4–6 — perception, attention, memory
- PAPER McClelland & Rumelhart (1981) — Interactive Activation Model
- READ PDP Vol 1 — Ch 1 (overview), Ch 8 (past tense)
- CODE Start building your chosen PDP model (word recognition, past tense, or categorization)

W28 The Systematicity Debate

- READ O'Reilly & Munakata Ch 7–9 — language, executive function, development
- PAPER Fodor & Pylyshyn (1988) — Connectionism and Cognitive Architecture — the core critique
- CODE Continue model development. Run experiments with mixed-effects analysis on results across seeds.

W29 Artifact

14–16 hrs

- CODE Analyze model representations (hidden unit activations, similarity structure, representational similarity analysis)

ARTIFACT DUE **PDP Modeling Project — includes response to Fodor & Pylyshyn**

Code + Report · 3,000 words + code

MODULE 2.4 · WEEKS 30–33

Computational Neuroscience

> *Mathematical models of neurons, circuits, and neural computation.*

W30 Neural Encoding

- READ Dayan & Abbott Ch 1–2 — neural encoding, firing rates, tuning curves
- WATCH Coursera Computational Neuroscience (UW) — Weeks 1–3
- CODE Simulate a leaky integrate-and-fire neuron. Plot firing rate curves.

W31 Neural Decoding and Information

- READ Dayan & Abbott Ch 3–4 — decoding, information theory, signal detection
- WATCH Coursera — Weeks 4–6
- CODE Implement a population coding model (e.g., orientation tuning in V1) with a Bayesian decoder

W32 Network Models

- READ Dayan & Abbott Ch 7–8 — network models, associative memory, attractor dynamics
- WATCH Coursera — Weeks 7–8
- WATCH Browse: MIT BCS Computational Tutorial Series — relevant tutorials
- CODE Start implementing your circuit model (drift-diffusion, Hopfield, or winner-take-all)

W33
14–16 hrs

Artifact

CODE Complete model. Analyze dynamics. Generate figures.

NOTE *Semester 2 review: you can now model cognition computationally at multiple levels with honest statistics. Take a breath.*

ARTIFACT DUE **Neural Circuit Model Report**

Code + Report · 2,500 words + code + figures

SEMESTER 3 · WEEKS 34–47

Brain, Behaviour, and Development

Memory, value, development, and language — the empirical heart of the field.

MODULE 3.1 · WEEKS 34–36

Memory and Learning Systems

> *What are the memory systems, how are traces formed and reformed, and why does consolidation never really end?*

W34 The Classical Taxonomy

- PAPER Tulving (1972) — Episodic and semantic memory
- PAPER Squire (1992) — Memory and the hippocampus: A synthesis from findings with rats, monkeys, and humans
- READ Eichenbaum — Cognitive Neuroscience of Memory Ch 1–4 (systems, hippocampal function, relational memory)

W35 Circuits, Oscillations, Reconsolidation

- READ Buzsáki — Rhythms of the Brain Ch 8 & 11 (hippocampal theta, sequence replay, memory consolidation)
- READ Eichenbaum Ch 5–8 (systems-level consolidation, pattern separation and completion)
- PAPER Nader, Schafe & LeDoux (2000) — Fear memories require protein synthesis in the amygdala for reconsolidation
- PAPER Dudai (2012) — The Restless Engram: Consolidations Never End

W36
12–14 hrs

Artifact

NOTE *A self-directed cog-sci programme that ignores memory is building on sand. This module closes that gap.*

ARTIFACT DUE **Memory Systems Essay — standard model of consolidation vs. reconsolidation: where each is right, where the evidence forces revision**

Written · 3,500 words

MODULE 3.2 · WEEKS 37–40

Reinforcement Learning and Decision Neuroscience

> *How do brains learn to value and to choose? The single most empirically productive arm of contemporary cog-neuro.*

W37

RL Foundations

READ Sutton & Barto — Reinforcement Learning: An Introduction Ch 1–4 (MDPs, dynamic programming, Monte Carlo)

PAPER Rescorla & Wagner (1972) — A theory of Pavlovian conditioning

W38

Dopamine as Prediction Error

READ Sutton & Barto Ch 5–7 (TD learning, n-step, planning)

PAPER Schultz, Dayan & Montague (1997) — A neural substrate of prediction and reward

PAPER Niv (2009) — Reinforcement learning in the brain

W39 Model-Based, Model-Free, and Decision Neuroscience

- READ Sutton & Barto Ch 8 (planning and learning with tabular methods)
- PAPER Daw, Gershman, Seymour, Dayan & Dolan (2011) — Model-based influences on humans' choices and striatal prediction errors
- READ Gershman — What Makes Us Smart Ch 3, 5, 7
- PAPER Gershman, Horvitz & Tenenbaum (2015) — Computational rationality

W40 Artifact
14–16 hrs

- CODE Implement the Daw two-step task and fit a hybrid model-based / model-free learner
- CODE Show how the mixing weight shifts under cognitive load or stress (use a published dataset; mixed-effects estimation per Module 1.5)

ARTIFACT DUE **Two-Step Task Implementation + Hybrid MB/MF Analysis**

Code + Report · 3,000 words + code + figures

MODULE 3.3 · WEEKS 41–44

Developmental, Comparative & Social Cognition

> *How does cognition come into existence — across development, across species, and between minds?*

W41 Development I — Piaget, Vygotsky, Core Knowledge

- READ Piaget — The Construction of Reality in the Child (selected chapters)
- READ Vygotsky — Mind in Society (selected chapters on the zone of proximal development)
- READ Spelke — What Babies Know Ch 1–5 (core knowledge of objects, agents, number, places)
- PAPER Spelke & Kinzler (2007) — Core knowledge

W42 Development II — Conceptual Change, Bayesian Theory-Theory, Number

- READ Carey — The Origin of Concepts Ch 1–4, 8 (bootstrapping and conceptual change)
- PAPER Gopnik & Wellman (2012) — Reconstructing constructivism: causal models, Bayesian learning mechanisms, and the theory-theory
- PAPER Xu & Garcia (2008) — Intuitive statistics by 8-month-old infants
- PAPER Dehaene, Izard, Spelke & Pica (2008) — Log or linear? Distinct intuitions of the number scale

W43 Comparative Cognition

- READ de Waal — Are We Smart Enough to Know How Smart Animals Are? Ch 1–5, 9–11
- PAPER Premack & Woodruff (1978) — Does the chimpanzee have a theory of mind?
- PAPER Call & Tomasello (2008) — Does the chimpanzee have a theory of mind? 30 years later
- PAPER Hauser, Chomsky & Fitch (2002) — The faculty of language: what is it, who has it, and how did it evolve?
- PAPER Santos & Rosati (2015) — The evolutionary roots of human decision making

W44
16–18 hrs

Social Cognition, Theory of Mind, WEIRD + Artifact

- PAPER Leslie (1987) — Pretense and representation: The origins of 'theory of mind'
- PAPER Baron-Cohen, Leslie & Frith (1985) — Does the autistic child have a 'theory of mind'?
- PAPER Wellman, Cross & Watson (2001) — Meta-analysis of theory-of-mind development
- PAPER Apperly & Butterfill (2009) — Do humans have two systems to track beliefs and belief-like states?
- PAPER Saxe & Kanwisher (2003) — People thinking about thinking people (rTPJ)
- PAPER Hickok (2009) — Eight problems for the mirror neuron theory of action understanding
- PAPER Sperber & Mercier (2011) — Why do humans reason? (argumentative theory)
- PAPER Henrich, Heine & Norenzayan (2010) — The weirdest people in the world?

ARTIFACT DUE **Uniquely-Human Cognition Essay — what (if anything) is uniquely human? Argue across core-knowledge, comparative, ToM, and cultural evolution evidence**

Written · 5,000 words

MODULE 3.4 · WEEKS 45–47

Language and Psycholinguistics

> *How does language relate to thought? Is it symbolic computation, embodied simulation, dialogic alignment, or something else?*

W45 Architecture of the Language Faculty

- READ Jackendoff — Foundations of Language Ch 1–6 (parallel architecture)
- PAPER Chomsky (1959) — Review of Skinner (historical context)
- PAPER Pinker & Jackendoff (2005) — The faculty of language: What's special about it?

W46 Sentence Processing, Alignment, Acquisition, Embodiment

- READ Bergen — Louder than Words (full book — accessible, engaging)
- READ Lakoff & Johnson — Metaphors We Live By (full book, ~190 pages)
- PAPER Pickering & Garrod (2004) — Toward a mechanistic psychology of dialogue
- PAPER Trueswell, Tanenhaus & Garnsey (1994) — Semantic influences on parsing
- PAPER Tanenhaus, Spivey-Knowlton, Eberhard & Sedivy (1995) — Integration of visual and linguistic information in spoken language comprehension
- PAPER Tomasello (2003) — Constructing a Language (selected chapters)

W47
12–14 hrs

Artifact

NOTE *Semester 3 review: you now have the empirical substrate — memory, value, development, language — that the embodied/phenomenological turn in Semester 4 will push against.*

ARTIFACT DUE

Language & Embodiment Analysis — embodied simulation, dialogic alignment, and the Chomskyan/Tomasello debate through computational + phenomenological lenses

Written · 3,500 words

SEMESTER 4 · WEEKS 48–61

The Embodied & Phenomenological Turn

Challenge the computational orthodoxy. What if the body, the world, and lived experience are constitutive of mind?

MODULE 4.1 · WEEKS 48–51

Embodied Cognition and Enactivism

> *Cognition isn't just in the head. It's in the body, the environment, and the dynamic coupling between them.*

W48 The Embodied Mind

- READ Varela, Thompson & Rosch — The Embodied Mind (full book, ~280 pages)
- PAPER Gibson (1979) — Chapter on affordances
- NOTE *This will challenge the computational framing you've built in Semester 2. Sit with the tension.*

W49 Situated Cognition

- READ Clark — Being There (full book)
- PAPER Brooks (1991) — Intelligence Without Representation
- PAPER Clark & Chalmers (1998) — The Extended Mind

W50 Deep Enactivism

- READ Thompson — Mind in Life Ch 1–4, Ch 7–8 (autonomy, sense-making, life-mind continuity)
- PAPER Di Paolo, Buhrmann & Barandiaran (2017) — Sensorimotor Life: selected chapters
- WRITE Start outlining your 5,000-word comparative essay

W51
16–18 hrs

Artifact

NOTE *This is a major essay. Block out extra time this week.*

ARTIFACT DUE **Comparative Framework Essay — computationalism vs embodied vs enactive**

Written · 5,000 words

MODULE 4.2 · WEEKS 52–54

Phenomenology for Cognitive Science

> *What can first-person experience teach us about the structure of the mind — and what does Dreyfus force computationalism to concede?*

W52

The Phenomenological Method

READ Gallagher & Zahavi — The Phenomenological Mind Ch 1–5 (method, intentionality, consciousness, self-consciousness)

READ Merleau-Ponty — Phenomenology of Perception, Preface + Part I Ch 1–3 (body schema, motor intentionality) — required, not optional

READ Noë — Action in Perception Ch 1–3 (perception as something we do)

W53

Body, Time, Intersubjectivity, and the Dreyfus Bridge

READ Gallagher & Zahavi Ch 6–10 — embodiment, time-consciousness, intersubjectivity, social cognition

READ Dreyfus — What Computers Still Can't Do (Introduction to the MIT edition + Ch 1, 7) — the essential bridge text against classical AI

PAPER Dreyfus (2007) — Why Heideggerian AI failed and how fixing it would require making it more Heideggerian

READ Optional: Gallagher — How the Body Shapes the Mind Ch 1–4

W54
14–16 hrs

Artifact

ARTIFACT DUE

Phenomenological Analysis — first-person description of a cognitive phenomenon + constraints on computational models, with Dreyfusian critique

Written · 4,000 words

MODULE 4.3 · WEEKS 55–58

Consciousness Studies

The hard problem, the easy problems, and the theories trying to bridge them — including the one most empirically grounded theory the 2010s canon tends to forget.

W55 The Hard Problem

READ Chalmers — The Conscious Mind Part I (Ch 1–5: the hard problem, supervenience, logical possibility of zombies)

PAPER Chalmers (1995) — Facing Up to the Problem of Consciousness

PAPER Block (1995) — On a Confusion About a Function of Consciousness (access vs phenomenal)

W56 Global Workspace Theory — Baars & Dehaene

READ Baars — A Cognitive Theory of Consciousness Ch 1–4 (origin of GWT)

READ Dehaene — Consciousness and the Brain Ch 1–5 (Global Neuronal Workspace, signatures of conscious access)

PAPER Dehaene & Changeux (2011) — Experimental and theoretical approaches to conscious processing

W57 **Competing Theories — IIT, HOT, PP, and Dennett**

- READ Koch — The Feeling of Life Itself Ch 1–5 (Integrated Information Theory)
- READ Dennett — Consciousness Explained Ch 5–7, 10–12 (multiple drafts model, qualia disqualified)
- READ Hohwy — The Predictive Mind Ch 1–3
- PAPER Lau & Rosenthal (2011) — Empirical support for higher-order theories
- PAPER Block (2007) — Consciousness, accessibility, and the mesh between psychology and neuroscience

W58 14–16 hrs **Artifact**

ARTIFACT DUE **Consciousness Theory Evaluation — contrast GWT, IIT, HOT, PP on empirical and conceptual grounds; does any solve the hard problem?**

Written · 4,500 words

MODULE 4.4 · WEEKS 59–61

Predictive Processing and Free Energy

> *The brain as a prediction machine — perception, action, and learning as inference.*

W59 **The Prediction Machine**

- READ Clark — Surfing Uncertainty Ch 1–5 (prediction machine, perception as controlled hallucination, action-oriented PP)
- READ Hohwy — The Predictive Mind Ch 4–8

W60 The Free Energy Principle

READ Clark — Surfing Uncertainty Ch 6–10 (attention, emotion, the hard problem through PP)

PAPER Friston (2010) — The Free-Energy Principle: A Unified Brain Theory?

PAPER Friston (2009) — Predictive Coding Under the Free-Energy Principle

PAPER Parr, Pezzulo & Friston (2022) — Active Inference: selected chapters

W61 Building the Model + Artifact

14–16 hrs

CODE Implement a simple predictive coding model — hierarchical message-passing between 2–3 layers

CODE Demonstrate prediction error minimization on a toy perceptual task (line orientation, simple pattern completion)

CODE Vary precision weighting; show how it changes model behavior

NOTE *Semester 4 review: you've traveled from computation to embodiment to phenomenology to prediction. The synthesis is coming.*

ARTIFACT DUE **Predictive Coding Model + Paper — connect computation to philosophy of perception and agency**

Code + Report · 3,000 words + code

SEMESTER 5 · WEEKS 62–80

Integration & Capstone

Catch up to 2026, pull everything together, ship original research.

MODULE 5.1 · WEEKS 62–64

Contemporary Computational Cognitive Science

> *Where is the field in the mid-2020s? Probabilistic programs, program synthesis, LLMs as cognitive models, and mechanistic interpretability.*

W62 Human-Machine Learning Comparison

- PAPER Lake, Ullman, Tenenbaum & Gershman (2017) — Building machines that learn and think like people
- PAPER Marcus (2018) — Deep learning: A critical appraisal
- PAPER Shiffrin & Mitchell (2023) — Probing the psychology of AI models

W63 Probabilistic Programs, Pragmatics, LLMs as Cognitive Models

- PAPER Frank & Goodman (2012) — Predicting pragmatic reasoning in language games (RSA)
- PAPER Goodman & Frank (2016) — Pragmatic language interpretation as probabilistic inference
- PAPER Ellis et al. (2021) — DreamCoder: Bootstrapping inductive program synthesis with wake-sleep library learning
- PAPER Binz & Schulz (2023) — Using cognitive psychology to understand GPT-3; Turning LLMs into cognitive models

W64
14–16 hrs

Mechanistic Interpretability + Artifact

PAPER Olah et al. (2020) — Zoom In: An Introduction to Circuits

PAPER Bricken et al. (2023) — Towards Monosemanticity (Anthropic)

PAPER Elhage et al. (2022) — Toy Models of Superposition

PAPER Sharkey et al. (2025) — Open Problems in Mechanistic Interpretability

CODE Reproduce a small mech-interp result — probe for induction heads on a small transformer using TransformerLens, or replicate one of the toy-models findings

ARTIFACT DUE **Contemporary Computational Cog-Sci Essay — are LLMs cognitive models? What counts as understanding in 2026? (with reproducibility notebook)**

Code + Report · 4,000 words + code

MODULE 5.2 · WEEKS 65–67

Cognitive Architectures

Can we build a unified theory of the mind? What would it take?

W65

Classical Architectures

READ Newell — Unified Theories of Cognition Ch 1–4 (the case for unified theories, Soar)

READ Anderson — How Can the Human Mind Occur in the Physical Universe? Ch 1–4 (ACT-R)

W66

Beyond Classical

READ Clark — Supersizing the Mind (full book: extended mind, cognitive assembly, scaffolded cognition)

NOTE *Revisit key chapters from Varela et al., Clark Surfing Uncertainty, Thompson Mind in Life, Lake et al. — now with architecture eyes.*

W67
16–18 hrs

Artifact

NOTE *The convergent essay where everything you've learned comes together — the intellectual capstone before the research project.*

ARTIFACT DUE **Cognitive Architectures Analysis — classical (ACT-R, Soar), connectionist, Bayesian, predictive processing, enactive, LLM-as-architecture. Propose criteria for a complete theory of mind.**

Written · 5,000 words

MODULE 5.3 · WEEKS 68–70

Dynamical Systems

> *Cognition as a dynamical process, not a computational one.*

W68

Foundations — Mind as Motion

READ Port & van Gelder (eds) — Mind as Motion Ch 1–3 (the dynamical hypothesis) + one applied chapter (decision making or speech)

READ Kelso — Dynamic Patterns Ch 1–4 (coordination dynamics, HKB model, metastability)

PAPER Beer (2000) — Dynamical Approaches to Cognitive Science

DO Work through basic dynamical systems math — phase spaces, attractors, bifurcations (3Blue1Brown + Strogatz Nonlinear Dynamics Ch 1–3 selections)

W69 Modeling Dynamics

- READ Hutchins — Cognition in the Wild Ch 1–4, Ch 9 (distributed cognition in naval navigation)
- READ Port & van Gelder — chapter on Smith & Thelen on A-not-B
- CODE Implement a dynamical systems model — HKB coupled oscillator, A-not-B (Thelen/Smith), or a drift-diffusion with collapsing bounds
- CODE Generate phase space plots, identify attractor states, demonstrate bifurcations

W70 Artifact
12–14 hrs

ARTIFACT DUE **Dynamical Systems Model + Report**
Code + Report · 2,500 words + code + phase space plots

MODULE 5.4 · WEEKS 71–80

Masters Research Project

> *The capstone. Original research integrating multiple strands of the curriculum — and the first piece of your work that lives or dies by external feedback.*

W71 Topic Selection and Scoping

- DO Decide your format: Edinburgh (10–15K dissertation), Stanford (qualifying paper + thesis), or single long-form paper
- WRITE Finalize topic. Suggested: "Cognitive Architecture as Lived Structure: A Predictive Processing and Enactive Account with Implications for AI Agent Design"
- WRITE Write: 1-page project proposal — research question, method, expected contribution, outline
- DO Identify 2 PhD students or early-career postdocs in the field who will read two drafts each for $200 per reader. Budget: $400–800 total. Reach out now; lead time is weeks, not days.

W72 Literature Review Deep Dive

READ Read 15–20 papers directly relevant to your topic (Google Scholar, Semantic Scholar, Connected Papers)

WRITE Write: Literature review draft (3,000–4,000 words)

NOTE *Identify the specific gap your project addresses.*

W73 Literature Review Refinement

WRITE Revise literature review based on gaps found while reading

READ Read 5–10 more papers to fill specific holes

WRITE Write: Refined literature review (4,000–5,000 words)

W74 Core Argument / Experiment I

WRITE Theoretical: develop your central argument — first main section. Each section advances a distinct claim supported by evidence.

CODE Empirical: begin designing and implementing your computational experiment — with pre-registered analysis plan (Module 1.5).

W75 Core Argument / Experiment II — First Paid Read

WRITE Theoretical: continue with second and third main sections.

CODE Empirical: collect results, iterate on design.

DO Send proposal + literature review + partial draft to Reader 1. Turnaround: 1–2 weeks.

W76 Drafting I

WRITE Full first draft — aim for ~50% of target word count

NOTE *Include: introduction, literature review, main body (argument or experiment + results), discussion, conclusion. Don't self-edit yet.*

W77 Drafting II — Integrate First Reader Feedback

WRITE Integrate Reader 1 feedback. Push to 80% of target word count.

NOTE *Get it all down first, then integrate structural feedback. Editing is the next phase.*

W78 Revision Pass 1 — Structure + Second Paid Read

WRITE Reread the full draft. Fix structural issues: does the argument flow? Are sections in the right order? Is the research question clearly answered?

WRITE Cut anything that doesn't serve the argument. Add where the argument is thin.

DO Send revised full draft to Reader 2 (fresh eyes). Turnaround: 1–2 weeks.

W79 Revision Pass 2 — Quality

WRITE Line-edit for clarity, precision, and academic voice

DO Check: every claim has a citation or argument. Every section connects to the next.

WRITE Integrate Reader 2 feedback. Optional: loop back to Reader 1 for a final short read of the discussion section.

W80 Final Revision, Presentation, and Completion

WRITE Final polish: abstract, bibliography, formatting

DO Prepare a 20-minute presentation of your research (slides or outline — even without audience, the exercise forces synthesis). Record yourself giving the presentation.

NOTE *Compile your full artifact portfolio — 21 pieces of work across ~18 months.*

NOTE *Celebrate. You just completed an MSc-equivalent education in cognitive science — with a methods spine, a real feedback loop, and contemporary reach.*

ARTIFACT DUE **Masters Research Project**

Dissertation · 10,000–20,000 words (format-dependent)

§ 04 Milestones

- **W03** Semester 1 started — first essay done — *Paradigm Map Essay* 1 / 21
- **W07** Philosophy of Mind complete — *Critical Essay on the Computational Theory of Mind* 2 / 21
- **W10** Cognitive Psychology complete — *Experiment Design Document (pre-registration style)* 3 / 21
- **W14** Neuroscience & Imaging Methods complete — *Brain–Cognition–Methods Map — annotated diagram linking brain structures to cognitive functions AND to the imaging modality that licenses each claim* 4 / 21
- **W17** **Semester 1 complete — foundations + stats spine built** — *Reanalysis of a Published Dataset — Bayesian hierarchical model + multiverse analysis + write-up with pre-registered analytic choices* 5 / 21
- **W21** First computational model built — *Three-Level Model Analysis* 6 / 21
- **W25** Bayesian modeling complete — *Bayesian Cognitive Model Report* 7 / 21
- **W29** Connectionist modeling complete — *PDP Modeling Project — includes response to Fodor & Pylyshyn* 8 / 21
- **W33** **Semester 2 complete — can model cognition computationally** — *Neural Circuit Model Report* 9 / 21
- **W36** Memory systems module complete — *Memory Systems Essay — standard model of consolidation vs. reconsolidation: where each is right, where the evidence forces revision* 10 / 21
- **W40** RL & decision neuroscience complete — *Two-Step Task Implementation + Hybrid MB/MF Analysis* 11 / 21
- **W44** Developmental + comparative + social essay done — *Uniquely-Human Cognition Essay — what (if anything) is uniquely human? Argue across core-knowledge, comparative, ToM, and cultural evolution evidence* 12 / 21
- **W47** **Semester 3 complete — brain + behaviour + development integrated** — *Language & Embodiment Analysis — embodied simulation, dialogic alignment, and the Chomskyan/Tomasello debate through computational + phenomenological lenses* 13 / 21

W51	Embodied cognition comparative essay done — *Comparative Framework Essay — computationalism vs embodied vs enactive*	14 / 21
W54	Phenomenology module complete — *Phenomenological Analysis — first-person description of a cognitive phenomenon + constraints on computational models, with Dreyfusian critique*	15 / 21
W58	Consciousness studies complete — *Consciousness Theory Evaluation — contrast GWT, IIT, HOT, PP on empirical and conceptual grounds; does any solve the hard problem?*	16 / 21
W61	**Semester 4 complete — embodied + phenomenological turn integrated** — *Predictive Coding Model + Paper — connect computation to philosophy of perception and agency*	17 / 21
W64	Contemporary computational cog-sci module complete — *Contemporary Computational Cog-Sci Essay — are LLMs cognitive models? What counts as understanding in 2026? (with reproducibility notebook)*	18 / 21
W67	Cognitive architectures capstone essay — *Cognitive Architectures Analysis — classical (ACT-R, Soar), connectionist, Bayesian, predictive processing, enactive, LLM-as-architecture. Propose criteria for a complete theory of mind.*	19 / 21
W70	Dynamical systems complete — *Dynamical Systems Model + Report*	20 / 21
W80	**Programme complete — dissertation submitted** — *Masters Research Project*	21 / 21

§ 05 Sources

This curriculum was synthesised from the course structures, reading lists, and assessment methods of:

- MIT Brain and Cognitive Sciences
- Stanford Symbolic Systems
- UC San Diego Cognitive Science
- University of Sussex Centre for Cognitive Science
- Carnegie Mellon Psychology (Cognitive Science track)
- University of Edinburgh Informatics
- Indiana University Cognitive Science

80 weeks · 21 artifacts · 25+ canonical textbooks · 60+ foundational papers · one self-made cognitive scientist.

Progress is committed to git — every checkmark reflects work shipped and a commit made.

FPL0_